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What drives green brand equity? Insights from the Delphi method and analytic hierarchy process

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Abstract

Green brand equity (GBE) is an emerging topic that is attracting growing attention from both scholars and managers. However, most empirical research in this field has predominantly employed quantitative methods. This study aims to identify the factors and criteria driving GBE ranking the relative importance of these factors and criteria. The Delphi method and the Analytic Hierarchy Process (AHP) were combined to address research gaps. The participants included individuals with at least five years of experience in consuming green agricultural products in Vietnam. The research results indicate the relative importance of the factors for GBE from high to low as follows: Subjective norms, emotional perceived values, functional perceived values, green trust, corporate social responsibility, attitude toward green products, and perceived behaviour control. The study also provided 24 criteria related to these factors and demonstrated their weights concerning GBE. This could potentially be the first study examining the relationship of all components within the theory of planned behaviour with GBE while ranking their relative importance compared to CSR and perceived value.

Keywords Green brand, Green brand equity, Brand management, Corporate social responsibility, CSR, Perceived value

1 Introduction

Growing environmental concerns have ushered in the era of “green strategy”, pushing sustainability issues to the forefront of numerous corporate agendas [1]. Companies are facing increased public sensitivity, stricter regulations, and growing pressure from stakeholders regarding responsible business practices and natural environment conservation [2]. Therefore, companies and governments are increasingly focusing on integrating environmental responsibility into their marketing and brand management strategies [3]. There is evidence suggesting that consumers have a positive attitude toward products and brands perceived to have little or no negative impact on the environment [4–6]. The literature has also provided evidence that customers are willing to pay a premium for brands that implement environmentally friendly policies [5].



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In this context, companies are actively adjusting their brand equity management strategies towards green brand equity (GBE) [6]. GBE is a recent trend that has been attracting the attention of researchers, managers, and policymakers. Several studies have shown the significant importance of GBE, which can positively influence consumers in various ways, such as positive word-of-mouth [7], brand attitudes [6], purchase intentions [8], and even brand passion [4].

There have been several studies on the factors driving GBE across different industries and contexts. For instance, [9] studied the electronics industry in South Korea, [10] focused on the cosmetics industry in Taiwan ROC, [6] examined shopping malls in India, [11] explored refrigerators and paper towels in Turkey, [12] analyzed telecommunications and home appliances in Pakistan and Italy, [13] investigated hotels in India, [14] researched fast-food restaurants in Pakistan, [15] explored banks in India, and [16] looked into the banking sector in Vietnam. These studies predominantly employed quantitative methods such as Structural Equation Modeling (SEM) and regression. They emphasize drivers like brand image, attachment, satisfaction, trust, loyalty, perceived quality, brand awareness, and brand associations. Some studies have focused on less commonly examined factors, such as green perceived value [13] or corporate social responsibility [16], but have not compared their relative importance with other dimensions of GBE. Moreover, these studies primarily employed the SEM method and concentrated on sectors other than agricultural products.

Agricultural products are among Vietnam's key and well-known export sectors, contributing significantly to the country's GDP and overall export turnover. However, according to Vietnam's Ministry of Industry and Trade [17], agriculture is also one of the main contributors to climate change, accounting for up to 18% of the nation's total greenhouse gas emissions. Emissions from Vietnam's agricultural sector are projected to reach 120 million tons of CO₂ by 2030. To fulfill its commitment at the 26th Conference of the Parties to the United Nations Framework Convention on Climate Change (COP26) to achieve net-zero emissions by 2050, greening the agricultural sector has become an urgent priority. Therefore, it is crucial to research effective solutions for enhancing GBE in Vietnam's agricultural industry.

The existing research on GBE so far has certain notable gaps. Firstly, there appears to be a lack of studies examining all the components of the Theory of Planned Behaviour (TPB) as a foundation to explain the drivers behind GBE development. Most studies to date have primarily focused on elements of GBE based on the definition of consumer-based brand equity of Keller [18] or explored the impact of marketing-mix factors on the structure of GBE. However, there is evidence suggesting that GBE is constituted by multiple factors from both the brand and individual consumer perspectives [19]. From a perceived benefit standpoint, consumers' intention to purchase green products might stem from within themselves, not solely from stimuli provided by businesses through marketing activities [20]. Secondly, there appears to be a lack of qualitative studies or a combination of qualitative and quantitative approaches to provide more insights than what conventional quantitative methods can typically explain. Particularly, there is an apparent absence of studies integrating Delphi and Analytic Hierarchy Process, and deriving GBE drivers based on the TPB structure. Despite the widely acknowledged role of perceived value types in general marketing research [13], studies that specifically examine the separate impacts of functional and emotional values on GBE, and compare

them within a green context, remain very limited. Furthermore, there seems to be a scarcity of studies examining consumers in the agricultural sector, despite it being a pivotal industry, consumed daily, and significantly impacting the environment.

This study aims to address the aforementioned gaps. It seeks to identify the factors and criteria that drive GBE, ranking the level of importance of these factors and criteria. Data were collected from consumers of green agricultural products in Vietnam. The Delphi and Analytic Hierarchy Process methods were combined to gather and analyze the data. The subsequent sections of the paper include the second section on theoretical background, the third section on methodology, the fourth section on research results, the fifth section on discussions, and the final section on conclusions.

2 Theoretical background

2.1 Green brand equity

Brand equity can be considered from either a financial or consumer standpoint. When viewed from the consumer's perspective, it is referred to as consumer-based brand equity [21]. Under this approach, brand equity is defined as the differentiated effects of brand knowledge on consumer perception through brand marketing activities [18]. It reflects the differences in customer choice between branded and non-branded products [22].

Green brand equity is viewed as the perceptions, beliefs, and understanding in the minds of consumers regarding a brand's sustainability and environmental friendliness [23]. GBE encompasses a set of perceptions and behaviors of customers towards a brand's environmental responsibility commitments, which elevate the brand's value [24]. Similar to brand equity, GBE refers to the added value gained by a brand. Consumers are believed to favor a product with a higher brand value over another with the same function. For green products, this added value relates to the product's attributes meeting consumers' environmental preservation needs. According to Bekk et al. [7], products with higher GBE are given priority by consumers. Studies indicate a close relationship between sustainable business practices and GBE [25].

2.2 Theory of planned behaviour

The Theory of Planned Behaviour by Ajzen [26] assumes that behavior can be predicted or explained by intentions to perform that behavior. This theory posits that intention is a function of three influencing factors: attitude toward the behavior, subjective norms, and perceived behavioral control.

Attitude toward the behavior refers to an individual's evaluation of the outcomes derived from performing a specific behavior, indicating the degree of favourability or unfavourability towards an individual's behavior [26]. Accordingly, environmentally conscious consumers are expected to hold a positive attitude towards green brands to meet their consumption objectives [19]. Some researchers have found that environmentally responsible consumers often exhibit positive attitudes towards environmentally friendly brands [21, 27, 28].

Subjective norms represent an individual's perception of the importance that significant others attach to whether the behavior should or should not be performed. Thus, an individual's intention and behavior can be influenced by the opinions of significant others, such as parents, spouses, friends, or colleagues [26]. Some studies related to GBE

have also suggested that positive word-of-mouth can impact visitation intentions, continued word-of-mouth, and willingness to pay a premium, aiding companies in attracting new consumers [29–31]. However, as of now, there seems to be a lack of experimental evidence showing that subjective norms drive intentions and behaviors toward consuming green products or supporting green brands.

Perceived behavioral control is an individual's perception of the ease or difficulty in performing a specific behavior, depending on the availability of resources and opportunities to carry out that behavior [26]. Therefore, it can be anticipated that human behavior and intentions toward consuming green products may be influenced by their perceptions of convenience, availability, opportunities, or challenges when purchasing those products. There is evidence indicating that establishing green distribution channels creates convenience for consumers, consequently positively impacting trust and loyalty toward green brands [2]. However, to date, the literature appears to provide very little empirical evidence on the relationship between perceived behavioral control and GBE.

Although widely applied in green consumer behavior research, TPB has faced several criticisms. A key limitation is the attitude-behavior gap, where consumers may hold positive attitudes toward green products but fail to act accordingly [31]. Additionally, components such as subjective norms and perceived behavioral control have shown inconsistent results in predicting actual behavior [32–34]. As a result, many studies have proposed extending TPB by incorporating factors like perceived value, willingness to pay, or green identity to enhance its predictive power [33]. Moreover, TPB needs to be contextually adapted, as cultural and social factors, such as material values or social media influence, can significantly alter green consumption intentions and behavior [35, 36].

2.3 Green trust, green perceived value

Green trust is the willingness to trust a product, service, or brand based on belief or expectation of its environmental effectiveness [23]. Several studies have demonstrated that trust in green brands is a fundamental driver of GBE [11, 17, 18]. Trust in the effectiveness of a green product regarding its environmentally friendly features and natural environment preservation is a key criterion for customers when purchasing it [23]. Therefore, green trust can strengthen the intention to purchase green products [35]. Furthermore, trust in a green brand indicates a higher likelihood that its customers will support it, and provide positive feedback, thereby enhancing GBE [36]. Papista et al. [5] identified the multidimensional drivers of GBE, including green trust, green brand image, green satisfaction, green brand experience, green customer value, and green marketing mix (7Ps). Among these, the role of green trust is particularly emphasized. When consumers believe that a brand is genuinely committed to environmental sustainability, it has a significant and positive impact on brand equity [37, 38].

Green perceived value is the comprehensive assessment of the extent to which consumers perceive their green needs to be met during an experience [39]. Customer perceived value is viewed as an exchange between perceived costs and perceived benefits [38]. Brands can offer both functional and emotional value to consumers [40]. Functional value refers to the practical and utilitarian benefits that a product provides, such as quality, performance, and cost-efficiency. It reflects how well a product fulfills its

intended purpose [40]. In contrast, emotional value captures the affective and psychological gratification derived from the product, such as enjoyment, comfort, or positive feelings evoked by its use. While functional value appeals to rational judgment, emotional value resonates with personal feelings and experiences [40]. These two dimensions often interact but remain conceptually distinct, each contributing uniquely to consumer decision-making. Li et al. [24], and Papista et al. [5] demonstrate that perceived value by consumers is highly significant in constructing and developing GBE. Therefore, creating greater value for customers will enhance the relationship between consumers and green brands, and conversely, customer reactions to green brands contribute to shaping GBE [24].

Although green perceived value stands as a crucial factor in the strategy for GBE development, the number of studies on this driver remains limited to date. Particularly, there appears to be a lack of research evaluating the impact of emotional values, and functional values on GBE while ranking their importance concerning other factors.

2.4 Corporate social responsibility

McWilliams and Siegel [41] define Corporate Social Responsibility (CSR) as actions that produce positive social outcomes and go beyond the interests of the firm and legal requirements. CSR is understood as a comprehensive construct encompassing multiple dimensions, including corporate citizenship, business ethics, sustainability, environmental responsibility, and both social and financial performance [42]. Rather than being merely philanthropic, CSR is increasingly viewed as a strategic move to enhance brand value by shaping social identity and encouraging consumer engagement [43]. Key elements such as a positive brand image and consumer trust, which are strengthened through CSR initiatives, serve as essential building blocks for sustainable brand value [44, 45].

CSR has a clear and positive impact on brand value in general—and particularly on GBE in the context of environmentally focused initiatives. Gilal et al. [4] found that consumer perceptions of CSR significantly boost brand trust and passion. CSR activities aimed at environmental sustainability, such as green advertising or zero-packaging initiatives, help increase brand awareness, trust, satisfaction, and loyalty, thereby strengthening green brand equity [44, 46]. Moreover, the relationship between CSR and GBE is mediated by factors like brand awareness, brand image, brand loyalty, and purchase intention [47]. These findings underscore CSR's role as a strategic tool in building and maintaining brand equity, especially when CSR is coherently integrated into the brand's core values and aligned with stakeholder expectations across global markets [48].

3 Methodology

3.1 Research design

This study utilizes a combination of the Delphi and Analytic Hierarchy Process (AHP) methods. The integration of these two methods is considered an effective way to make decisions in complex situations [49], aiding in reducing individual biases while enhancing the efficiency and transparency of the decision-making process [50]. It is also seen as the amalgamation of powerful tools for decision-making across various fields, including business, policy, and science [51]. The application of the Delphi and AHP methods is particularly appropriate given the complexity and exploratory nature of the research

topic. GBE is an emerging area with limited empirical evidence, especially regarding the relative importance of various influencing factors. The Delphi method enables systematic expert consultation to identify and refine key factors, while AHP allows for the structured comparison and prioritization of these factors based on participant judgment. Combining these methods not only enhances the validity of the findings by reducing individual bias but also improves the transparency and rigor of the decision-making process.

Figure 1 illustrates the research design of this study. Firstly, we examined relevant literature on GBE across different countries and industries. Subsequently, we identified gaps and specified the particular research objectives. Drivers that have been minimally explored or absent in the literature but have theoretical foundations or scientific bases were selected for examination. Then, we proceeded to develop specific criteria (within the factors), based on the TPB and the overall literature review.

Following that, we invited six experts to contribute their opinions on the criteria set to evaluate their suitability for the agricultural sector within the context of Vietnam. Among these experts, three were retail store managers specializing in the sale of green agricultural produce (acting both as consumers and sellers), while the remaining three were researchers and lecturers in the field of brand management, holding doctoral degrees. All experts possessed a minimum of five years' experience in consuming green

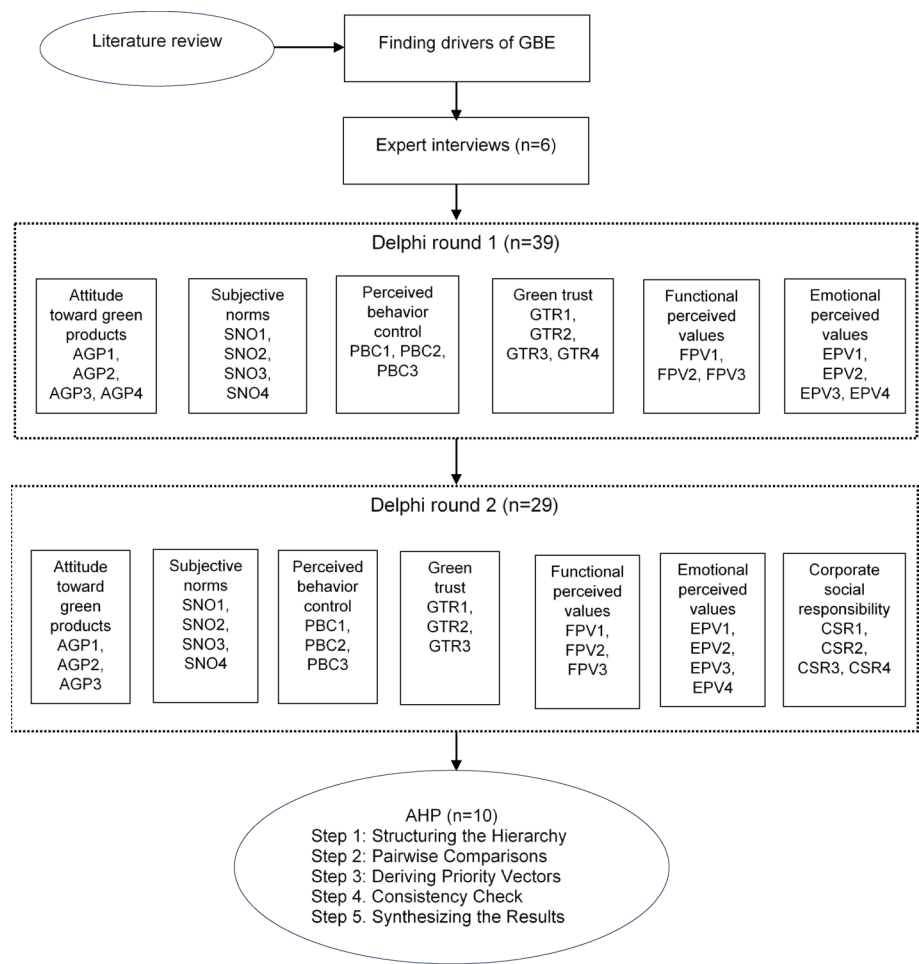


Fig. 1 Research design

agricultural produce. The initial questionnaire was formed based on adjustments made according to the participants' feedback.

Subsequently, the Delphi method was applied with two survey rounds. Finally, the AHP was employed to rank the importance level of the factors and criteria. Interviews and surveys were conducted in person from September to November 2023 in Hanoi, Vietnam. Detailed steps of the research are presented in Sects. 3.2 and 3.3.

Regarding the characteristics of the sample across the rounds, in Delphi Round 1, a total of 39 participants took part, including 32 females (82.1%) and seven males (17.9%). In Delphi Round 2, the number decreased to 29 participants, with 26 females (89.7%) and three males (10.3%). By the AHP Round, only 10 participants remained, consisting of eight females (80%) and two males (20%). Regarding age, in the first Delphi round, there were nine participants (23.1%) aged 18–30, 13 (33.3%) aged 31–50, and 17 (43.6%) aged 51 and above. In Delphi Round 2, the numbers were eight (27.6%) for both the 18–30 and 31–50 age groups, and 13 (44.8%) for those aged 51 and over. In the AHP Round, no participants were from the 18–30 age group, while four (40%) were aged 31–50 and five (50%) were aged 51 and above.

3.2 Delphi

The Delphi method is a technique introduced by Dalkey and Helmer [52] to gather opinions from a group of experts systematically on a specific issue to be resolved. Subsequently, Murry and Hammons [53] developed the modified Delphi method suggesting the use of structured questionnaires to focus participants on the issue being discussed and save time [54]. This study utilizes the modified Delphi method, allowing the researcher to begin with structured questions and also initiate some open-ended questions [55].

In the first Delphi round, the questionnaire comprised 25 questions. It began with an explanation of the GBE concept and a screening question to select survey participants, ensuring they have at least five years of experience in consuming green agricultural products. Following this were demographic questions. Then, questions about attitudes towards green products were adapted from [19], and questions about green trust were adapted from [9] and [23]. The remaining questions on social norms, perceived behavioral control, functional perceived values, and emotional perceived values were self-developed by the authors based on the TPB, Literature, and expert feedback. The structured questions were designed using a 5-point Likert scale, from highly important (5 points) to not important (1 point). Finally, there was an open-ended question for participants to provide further explanation and propose additional criteria for developing GBE.

Participants in the first Delphi round comprised 39 individuals. The mean value and Content Validity Ratio (CVR) will be used to analyze the data, assisted by Excel software. The CVR is an index used to assess the level of expert agreement regarding the necessity of each item or criterion. Each expert rates an item as “essential,” “useful but not essential,” or “not necessary.” According to Lawshe [56], CVR is calculated based on the number of experts who rate an item as “essential.” If the CVR value exceeds a specific threshold, depending on the number of experts (for example, a Minimum CVR of 0.31 is required for a panel of 35 or more), the item is considered to Have acceptable content

validity. With 39 participants in this study, the CVR threshold is 0.31. Criteria with a mean value < 3.5 or CRV < 0.31 will be eliminated [56].

In the second Delphi round, the questionnaire will be adjusted based on the results from the first round. There were 29 out of 39 (74%) participants continuing their involvement in the second round. With this sample size, the CVR threshold is 0.37 [56]. Data analysis will be conducted similarly to the first round. Some questions might be eliminated or modified. Additionally, based on the feedback obtained from participants through open-ended questions in the first round, new questions may be added to the questionnaire in the second Delphi round.

3.3 Analytic hierarchy process

To assess which factors drive GBE and their relative importance, numerous criteria are required for evaluation. Therefore, this evaluation process can be considered a multi-criteria decision-making process, and the application of the AHP technique is suitable [49]. According to Saaty [49], AHP has several benefits, including (1) focusing on determining the importance of each criterion, thus helping overcome shortcomings of many other methods and facilitating their integration to leverage the advantages of each method; (2) potentially enhancing reliability by checking consistency in experts' evaluation results; (3) being straightforward, considering both quantitative and qualitative factors, as well as both major and minor criteria.

When integrating Delphi with AHP, Khorramshahgol & Moustakis [57] recommend avoiding an excessive number of experts as interview subjects; typically, 5–15 people are most appropriate. We collected responses from ten participants who participated in the initial two rounds of Delphi for this AHP round. The questionnaire was designed based on the results obtained from the two Delphi rounds. The data were analyzed using MS Excel.

According to Saaty [49], AHP can be implemented through the following five steps:

3.3.1 Step 1: Structuring the Hierarchy

The first step involves constructing a hierarchical structure of the decision problem. This hierarchy typically includes the overall goal at the top level, followed by the evaluation criteria and sub-criteria, and finally the alternatives at the lowest level. The hierarchical decomposition helps decision-makers handle complex problems in a systematic way.

3.3.2 Step 2: Pairwise comparisons

In this step, decision-makers conduct pairwise comparisons of elements at each level of the hierarchy concerning their impact on the elements at the higher level. The fundamental 1–9 scale of Saaty [49] is used to represent the intensity of importance or preference. These judgments are used to construct a pairwise comparison matrix $A = [a_{ij}]$, where a_{ij} indicates the relative importance of element i compared to element j .

3.3.3 Step 3: Deriving priority vectors

To determine the relative weights of the elements, the pairwise comparison matrix is normalized, and the priority vector w is derived. This is typically done by calculating the principal right eigenvector of the matrix:

$$Aw = \lambda_{\max} w$$

where $\lambda_{\max}$ is the maximum eigenvalue of the matrix. The vector w represents the local priorities or relative importance of the elements.

3.3.4 Step 4: Consistency check

AHP incorporates a mechanism to assess the consistency of the pairwise comparisons. The Consistency Index (CI) is computed as:

$$CI = \frac{\lambda_{\max} - n}{n - 1}$$

where n is the number of elements compared. Then, the Consistency Ratio (CR) is calculated as:

$$CR = \frac{CI}{RI}$$

where RI is the average random consistency index, based on matrix size. A value of $CR \leq 0.1$ is generally considered acceptable, indicating that the judgments are reasonably consistent.

3.3.5 Step 5: Synthesizing the results

The final step is to synthesize the results to determine the global priorities of the alternatives. This involves aggregating the local weights from each level of the hierarchy to calculate the overall weight for each alternative:

$$\text{Overall Priority}_i = \sum_{j=1}^n w_j p_{ij}$$

where w_j is the weight of criterion j , and p_{ij} is the priority of alternative i under criterion j . The alternative with the highest overall priority is considered the most preferred choice.

The result obtained from integrating Delphi and AHP will be a set of factors, criteria driving GBE, and the resulting ranking of their relative importance.

4 Result

4.1 Delphi round 1

Table 1 presents the results from the first round of the Delphi survey. Two criteria, AGP4 and GTR4, were excluded due to a CVR index of < 0.31 . The criterion with the highest CVR is SNO1, followed by EPV1, and then AGP1, EPV3, and SNO3.

When answering the questions, some participants provided additional explanations, emphasizing the role of subjective norms. They expressed that awareness of the need for responsible consumption in emerging markets like Vietnam is still limited, thus advocating for enhanced education within families, educational institutions, and society to influence and exert “pressure” on each other toward sustainable consumption. Some participants highlighted the psychological benefits they experience when consuming green products. They feel secure and joyful, take pride in being smart consumers, and strongly desire green brands to cater to these needs.

Table 1 The results of Delphi rounds

Factors	Criteria		Round 1			Round 2		
			Average	CVR	Result	Average	CVR	Result
Attitude toward green products (AGP)	AGP1	Buying environmentally friendly products makes me feel good	3.97	0.49	Accepted	3.83	0.45	Accepted
	AGP2	I'm excited by the availability of environmentally friendly brands for the product categories I wish to buy	3.82	0.38	Accepted	3.86	0.38	Accepted
	AGP3	Among the several product categories, the brand that is favorable to the environment is always the best option	4.00	0.38	Accepted	3.97	0.38	Accepted
	AGP4	When I buy things from an environmentally friendly brand, I usually feel really satisfied	3.62	0.28	Rejected			
Subjective norms (SNO)	SNO1	My family encourages me to use green products	4.13	0.59	Accepted	4.07	0.66	Accepted
	SNO2	My friends encourage me to use green products	3.97	0.44	Accepted	4.03	0.52	Accepted
	SNO3	My colleagues encourage me to use green products	3.95	0.49	Accepted	3.97	0.52	Accepted
	SNO4	In general, the community around me encourages each other to use green products	3.79	0.33	Accepted	3.93	0.45	Accepted
Perceived behavior control (PBC)	PBC1	Finding and purchasing green products is easy for me	3.85	0.33	Accepted	3.72	0.38	Accepted
	PBC2	Regularly using green products is feasible for me	3.74	0.33	Accepted	3.79	0.38	Accepted
	PBC3	I know specific, necessary steps to buy green products from reputable green brands	3.82	0.38	Accepted	3.86	0.45	Accepted
Green trust (GTR)	GTR1	These brands engage in reliable community service	3.92	0.33	Accepted	4.00	0.38	Accepted
	GTR2	These brands are serious about protecting the environment	3.85	0.38	Accepted	3.86	0.45	Accepted
	GTR3	These brands fulfill their pledge to engage in environmental conservation efforts	3.79	0.33	Accepted	3.83	0.38	Accepted
	GTR4	These brands' pledges to assist their customers are incredibly trustworthy	3.79	0.18	Rejected			
Functional perceived values (FPV)	FPV1	Green features satisfy my needs	3.79	0.44	Accepted	3.83	0.45	Accepted
	FPV2	The benefits of the product have helped me protect the environment	3.87	0.33	Accepted	3.90	0.38	Accepted
	FPV3	The product's functionalities meet my expectations of environmental friendliness	3.87	0.33	Accepted	4.00	0.45	Accepted

Table 1 (continued)

Factors	Criteria		Round 1			Round 2		
			Average	CVR	Result	Average	CVR	Result
Emotional perceived values (EPV)	EPV1	These brands make me more confident about the health of my family	4.08	0.54	Accepted	4.10	0.52	Accepted
	EPV2	These brands make me more confident about the overall health of the community	4.03	0.44	Accepted	4.07	0.45	Accepted
	EPV3	These brands make me happy because they contribute to protecting the environment	3.95	0.49	Accepted	3.97	0.45	Accepted
	EPV4	These brands make me feel proud to be a conscientious consumer who cares about the environment	3.79	0.33	Accepted	3.97	0.38	Accepted
Corporate social responsibility (CSR)	CSR1	These brands often actively sponsor cases of social vulnerability				4.14	0.45	Accepted
	CSR2	These brands often actively participate in community development programs				4.07	0.45	Accepted
	CSR3	These brands often actively engage in environmental protection programs				4.07	0.38	Accepted
	CSR4	Overall, these brands have a good social responsibility				3.93	0.38	Accepted

Additionally, some participants mentioned that, apart from the factors already included in the questionnaire, attention should be given to CSR criteria. They believe that, in many instances, consumers only become aware of a brand's environmentally friendly characteristics through the company's CSR activities. They believe that companies demonstrating robust CSR practices will establish a positive image and trust among consumers, thereby increasing the intention to purchase green products and fostering GBE development.

4.2 Delphi round 2

Based on the feedback from participants in the first Delphi round and information regarding CSR, we have named this new factor CSR and introduced four new criteria, namely CSR1, CSR2, CSR3, and CSR4 for the second round. Therefore, after the first round, there are a total of seven factors and 24 criteria in the questionnaire, which include the remaining 20 criteria and four newly added criteria.

Table 1 illustrates that after the second Delphi round, all criteria Have gained consensus among the participants. The average values for all criteria are greater than 3.5, with CVR values surpassing 0.37. The new criteria introduced in round 2 have all been accepted. The criterion with the highest consensus based on the CVR index is SNO1, followed by EPV1, SNO2, SNO3, and SNO4, which belong to the subjective norms and emotional perceived values factors.

Table 2 Ranking of factors driving GBE

Factors	Weights	Ranking
Subjective norms	0.225	1
Emotional perceived values	0.168	2
Functional perceived values	0.144	3
Green trust	0.135	4
Corporate social responsibility	0.123	5
Attitude toward green products	0.107	6
Perceived behavior control	0.104	7
Consistency ratio	0.074	

Table 3 Weights and ranking of criteria driving GBE

Factors	Local weights	CR	Items	Local weights	Global weights	Ranking
Attitude toward green products	0.107	0.085	AGP1	0.329	0.035	15
			AGP2	0.264	0.028	21
			AGP3	0.407	0.043	10
Subjective norms	0.225	0.082	SNO1	0.358	0.081	1
			SNO2	0.287	0.065	2
			SNO3	0.214	0.048	7
			SNO4	0.142	0.032	17
Perceived behavior control	0.104	0.058	PBC1	0.440	0.046	9
			PBC2	0.255	0.027	22
			PBC3	0.305	0.032	18
Green trust	0.135	0.072	GTR1	0.399	0.054	5
			GTR2	0.383	0.052	6
			GTR3	0.218	0.029	20
Functional perceived values	0.144	0.084	FPV1	0.404	0.058	4
			FPV2	0.323	0.047	8
			FPV3	0.272	0.039	13
Emotional perceived values	0.168	0.070	EPV1	0.349	0.058	3
			EPV2	0.246	0.041	12
			EPV3	0.211	0.035	14
			EPV4	0.194	0.033	16
Corporate social responsibility	0.123	0.084	CSR1	0.345	0.042	11
			CSR2	0.254	0.031	19
			CSR3	0.213	0.026	23
			CSR4	0.188	0.023	24

4.3 Analytic hierarchy process

Table 2 presents the results of ranking the importance level of factors driving GBE and the consistency ratio of the assessment results from the participants. The CR is less than 0.1, ensuring consistency in the respondents' answers [49]. Thus, subjective norms, a component of TPB and seemingly unexplored in GBE literature, were evaluated as the most crucial factor. Following this are emotional perceived values, functional perceived values, green trust, corporate social responsibility, attitude toward green products, and perceived behaviour control, in descending order of importance.

Table 3 displays local weights, global weights, CR, and the ranking results of the criteria's importance. The five most crucial criteria are SNO1, SNO2, EPV1, FPV1, and GTR1, which belong to subjective norms, emotional perceived values, functional perceived values, and green trust factors. Although the new CSR criteria added in the second Delphi round are not ranked as high compared to other criteria, they still achieved consensus among the participants. The consistency ratio results indicate that all factors

fall below the acceptable threshold ($CR < 0.1$), ensuring the reliability of the pairwise comparison process in the AHP model [49]. Some factors, such as attitude toward green products (0.085), functional perceived values (0.084), and corporate social responsibility (0.084), show relatively higher CR values, which may be attributed to the large number of criteria within each group, making it cognitively demanding for experts, especially for more abstract constructs like CSR or technically complex ones like functional values. Nevertheless, these CR values remain within acceptable limits. In contrast, factors such as perceived behavioral control (0.058), emotional perceived values (0.070), and green trust (0.072) have lower CR values, reflecting a higher level of agreement among experts. This may be due to the clearer and more distinct nature of the criteria within these groups, which are commonly understood and consistently interpreted. Overall, these results strengthen the reliability of the AHP model's outcomes.

5 Discussion and implication

5.1 Discussion

This study contributes significantly to the theoretical aspects. Firstly, it might be pioneering in comprehensively examining the factors of the TPB (attitude toward green products, subjective norms, perceived behavior control) and demonstrating their crucial roles concerning GBE in comparison to other factors. Secondly, the ranking of the importance of various GBE-driving factors through the integration of the Delphi and AHP methods represents a rare approach to addressing gaps in the literature on GBE. This combination of qualitative and quantitative methods has provided additional information and explanations regarding the essential roles of subjective norms, emotional values, and corporate social responsibility in GBE development strategies. Thirdly, the distinct exploration of the relationship between emotional values, functional values, and GBE is a fresh approach compared to existing literature, highlighting and explaining the significance of each value type in consumer perceptions. Until now, there have been only a few experimental studies primarily employing linear structural models to explore the relationship between perceived value in general and GBE. Lastly, this is also one of the very few studies examining the relationship between CSR and GBE (not general brand equity).

Subjective norms emerged as the most important factor as ranked by the participants. The criteria under subjective norms, SNO1 (My family encourages me to use green products) and SNO2 (My friends encourage me to use green products), were ranked first and second, respectively. This finding is intriguing because surprisingly, despite the wide use of TPB to explain intention and behavior in business, there seems to be scarce experimental research examining the relationship between all three TPB components and GBE. This study has highlighted the significance of the opinions of reference groups such as family, friends, colleagues, and the public around each consumer. This can be understood in the context of Asian countries like Vietnam and the diversity of agricultural products, which are used daily and often rely heavily on word-of-mouth recommendations. These societies are known for their high-context nature and collectivism, where decision-making and actions are preferred based on the opinions of significant others.

Among the items of subjective norms, the influence of family and friends (SNO1, SNO2) emerged as the most influential in green consumption behavior, as they share

close personal relationships with the consumer. In contrast, the influence from the broader community (SNO4) has the lowest global weight (0.032), suggesting that social pressure from the general public has less impact. This may be attributed to the fact that in emerging markets with rapid economic growth and high consumption levels, such as Vietnam, public awareness of environmental protection remains limited, and the broader community has yet to significantly influence individuals' green consumption intentions.

Emotional perceived values and functional perceived values are the second and third most influential factors driving GBE, with the emotional criterion "These brands make me more confident about the health of my family" ranking third in global importance. This suggests that affective benefits, particularly those tied to consumer well-being and environmental reassurance, play a central role in shaping perceptions of green brands. In comparison, the functional dimension, as represented by "Green features satisfy my needs," ranked slightly lower but still reflects a strong utilitarian motivation.

Some prior studies have demonstrated a positive relationship between perceived values in general with GBE [13, 58, 59]. This study reinforces previous findings on the role of perceived value in GBE within the agricultural sector and the Vietnamese context, which differs from the industries and settings previously investigated, such as the hospitality sector in [13]. However, to date, there has been limited research isolating psychological and functional values for experimental evaluation. For instance, Winit and Kantabutra [60] discussed that psychological and functional benefits positively impact brand equity in general (not GBE). Sweeney and Soutar [40] positioned functional value as the primary driver in their PERVAL framework, particularly in contexts where product quality, performance, and price dominate decision-making. Emotional value, while acknowledged, was generally secondary to utilitarian concerns in their study. However, it is important to note that their work focused on broader consumer value judgments across product categories and not specifically within the green branding context. This study reveals a shift in emphasis within the green market, where emotional perceived values play a more dominant role than functional values in influencing green brand equity. This contrasts with traditional models like [40], which prioritize functional value and do not empirically address this hierarchy in a green branding context. In emerging markets such as Vietnam, where environmental concerns are increasingly pressing, consumers tend to value brands that offer not only functional benefits but also emotional experiences tied to health, safety, and pride. This highlights the need to adapt conventional value frameworks to the socio-environmental realities of sustainable consumption.

Green trust was ranked as the fourth factor. Although there have been numerous experimental pieces of evidence concerning the relationship between green trust and GBE, we aimed to pioneer in examining the relative importance of this factor in comparison to the components of TPB and perceived values. Prior studies have demonstrated that green trust has a positive, direct impact on GBE or serves as an intermediary in the relationship between attitude toward green sponsorship or GBE in the dairy industry [61], between CSR or green brand association, or green brand satisfaction with GBE in the banking sector [16]. The findings of this study further reinforce those earlier results, but in a new context: the agricultural sector in Vietnam. However, when compared with other factors like subjective norms and perceived values, green trust appears to

be relatively less influential and should be considered in conjunction with other antecedents of GBE. To effectively enhance green trust and GBE, priority should perhaps be given to strengthening subjective norms (through education, family, schools, and public advocacy) and perceived values (by improving customers' real-life experiences).

Corporate social responsibility was ranked fifth in importance among the seven factors. Similarly, its items, CSR1, CSR2, CSR3, and CSR4, were also ranked at moderately high levels, positioned at 11th, 19th, 23rd, and 24th respectively in the overall ranking. The role of CSR in GBE has been explored by a few recent studies [16, 61]. A unique aspect of our study is the integration of the CSR factor with other factors and their weighted rankings. The research results have indicated that CSR continues to play a significant role in the GBE development strategy of agricultural businesses. However, businesses should give greater priority to subjective norms, perceived values, and green trust, particularly in emerging markets like Vietnam, where public concerns about greenwashing, the practice of presenting products or activities as more environmentally friendly than they truly are, are increasing consumer skepticism [62]. Greenwashing has been shown to trigger negative brand perceptions and pose both economic and social risks [63, 64]. To address consumer skepticism toward this phenomenon, research suggests that companies must build green trust through transparency and honesty in their sustainability claims [64]. Organizations can also play a role in educating consumers about greenwashing and how to recognize genuine sustainability efforts, helping to reduce misunderstandings and foster trust [64]. In the context of a market showing signs of greenwashing in Vietnam, consumers might remain skeptical about corporate CSR-related activities, placing higher importance on green values and green trust. Nevertheless, the research findings still demonstrate that CSR is an indispensable activity for GBE development, which needs improvement concurrently with other driving factors.

Other components of the Theory of Planned Behavior, namely attitude toward green products and perceived behavioral control, were ranked as the least important factors (with local weights of 0.107 and 0.104, respectively). However, they still contribute to promoting GBE, as their consistency ratios (CR = 0.085 and 0.058) remain below the accepted threshold of 0.1 [49]. Moreover, in the second round of the Delphi process, all items under these two factors exceeded the Minimum required CVR threshold of 0.37, as recommended by Lawshe [56]. This indicates a reasonable level of expert consensus regarding their role. This result is partly supported by prior research. For instance, Ha [36] illustrated that attitude toward green products positively influences GBE in the electronics industry in Vietnam through intermediary factors like green brand image and green trust. Shanti and Joshi [13] demonstrated that a green attitude contributes to GBE in the hotel industry in India. Bekk et al. [7] indicated that brand attitude directly impacts positive word-of-mouth communication in Germany. However, there seems to be a lack of studies examining the relationship between perceived behavior control and GBE. Hence, this study contributes new knowledge to the GBE literature and demonstrates that consumers' ability to find and purchase green products easily and feasibly can promote GBE.

5.2 Implication

This study makes a theoretical contribution by empirically examining all three components of the TPB and comparing their relative importance to other drivers of GBE.

The integration of Delphi and AHP methods offers a rare mixed-method approach that enhances both qualitative and quantitative understanding. It also distinguishes between emotional and functional values, often merged in prior research, thus expanding the theoretical framework of perceived value in green branding. Moreover, it is among the few studies to explore the link between CSR and GBE rather than general brand equity. These findings suggest research implications: future studies should adopt longitudinal designs to capture the evolving influence of GBE drivers over time, and further investigate the distinct roles of emotional and functional values in shaping green consumer behavior, especially in emerging markets.

Based on these findings, the study offers several practical and policy implications aimed at enhancing GBE in emerging markets like Vietnam. First and foremost, there is a need to strengthen social norms through education and communication. Regulatory bodies should integrate sustainability and green consumption into national school curricula while launching large-scale awareness campaigns through schools, youth organizations, women's unions, and mainstream media. These channels have the potential to amplify pro-environmental social norms—identified in this study as the most influential factor in green purchasing behavior.

In addition, a transparent, credible, and accessible national green product certification system should be established. Such a system would enhance consumer trust in green brands while reducing concerns about greenwashing. For businesses, especially small enterprises and those in agriculture, investing in multiple GBE-related drivers may pose financial challenges. Therefore, governments should provide targeted support such as subsidies, tax incentives, and access to green financing for businesses adopting sustainable practices like cleaner production, biodegradable packaging, or eco-friendly supply chains. To ensure practicality, official cost–benefit assessment guidelines should be issued to help businesses prioritize investments in the most impactful drivers, rather than attempting to implement all seven simultaneously.

Moreover, to strengthen community-level impact, which was found to be relatively weak in this study, local campaigns such as “Green Consumption Week,” “Eco-friendly Farmers’ Markets,” or “Sustainable Brand Awards” should be organized regularly. These initiatives can help foster a collective green mindset and gradually elevate public awareness. Overall, the above actions require coordinated efforts among government agencies, businesses, civil society organizations, and local communities, with the long-term goal of building a sustainable consumption culture and improving the quality of life in Vietnam’s green growth context.

6 Conclusion

This study aims to identify the factors and criteria for developing GBE in the case of the agricultural sector in Vietnam, ranking the level of importance of these factors and criteria. The research results indicate the relative importance of factors in descending order as follows: Subjective norms, emotional perceived values, functional perceived values, green trust, corporate social responsibility, attitude toward green products, and perceived behavior control. The criteria within these factors have achieved consensus and have been ranked in terms of importance by the participants.

This work makes significant contributions to both science and practical application. It may be one of the few studies that examine the relationship between components of

the TPB and other factors with GBE, while also providing a weighted ranking of these factors. It might also be pioneering in combining the Delphi and AHP methods in GBE research, addressing certain research gaps, and offering new insights. This study has also attempted to explore factors rarely or never addressed in empirical GBE studies, such as emotional perceived values, functional perceived values, and CSR, providing profound insights and meaningful implications in this field. Moreover, the focus of this research is the agricultural sector, which is a pivotal economic sector in Vietnam and several other countries. It has a significant environmental impact but has been overlooked in prior GBE studies. Finally, the study offers practical recommendations for educational institutions, civil society organizations, corporations, and governments to improve GBE. Specific suggestions encompass the incorporation of sustainability and eco-friendly consumption into educational programs, the implementation of coordinated communication initiatives to enhance community awareness GBE, and the establishment of a national certification system for green products.

Certain limitations persist in this article. Although the sample sizes for the Delphi and AHP rounds meet methodological requirements, they remain relatively small and are geographically limited to Hanoi, which reduces the generalizability of the findings. The expert panel also lacks diversity, notably with no representation from environmental NGOs or farmers, potentially leading to a commercial bias. In addition, one of the main limitations of this study is that the CSR criteria introduced after the first Delphi round were not pilot-tested beforehand. Future studies might increase the number of participants in the Delphi and AHP phases, with a more diverse panel that includes representatives from various stakeholder groups such as environmental NGOs, farmers, policymakers, and consumers. Additionally, some other factors related to GBE were not examined in this study, such as green brand loyalty, brand associations, green brand image, and green satisfaction. Subsequent research could include new measures, especially considering the relationship between intention and behavior in purchasing green products, as TPB suggests that factors do not necessarily lead directly to action but often go through an intention phase. In addition, future scholars may also consider adopting longitudinal research designs instead of relying solely on cross-sectional data, as done in this study, to track how the influence of various factors on GBE evolves. This approach would help overcome the limitations of cross-sectional studies and offer deeper insights into the development of consumer attitudes and behaviors as sustainability initiatives unfold. Moreover, cross-national or cross-sector studies are encouraged to examine how cultural, economic, or industry-specific differences may shape the drivers of green brand equity, thereby enhancing the generalizability and contextual relevance of future findings. Finally, the scope of this study is focused on consumers of green agricultural products in Vietnam. Therefore, applying the research findings to other industries, fields, or countries requires caution. Future studies might explore GBE from the perspectives of other stakeholders, such as shareholders, and company employees, while also exploring other sectors significantly impacting the environment.

Author contributions

H.T.T wrote the main manuscript text; H.T.N prepared theoretical background and checked validation. All authors reviewed the manuscript.

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Data availability

The datasets analysed during the current study are available in the Dataverse repository, <https://doi.org/10.7910/DVN/NIBYE3>.

Declarations**Consent to participate**

Written informed consent was obtained from all participants prior to data collection. Participants were explicitly informed that their involvement was voluntary and that they retained the right to withdraw at any time. All procedures were designed to ensure the protection of participants, and interview transcripts were anonymized prior to being used solely for scientific purposes. No minors participated in the research.

Consent for publications

Not applicable.

Competing interests

The authors declare no competing interests.

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